Q9. Hypothesis Testing - Independent T-test

Original Aim: To determine whether the mean SepalLengthCm differs significantly between Iris-setosa and Iris-versicolor. State the null and alternative hypotheses, perform an independent two-sample t-test using SciPy, and display the t-statistic and p-value. Use a 0.05 significance level to state the conclusion.

Aim

Test whether population mean SepalLengthCm differs between Iris-setosa and Iris-versicolor using an independent two-sided t-test at 0.05 significance.

Requirements

Python, Pandas, scikit-learn and scipy.stats. Reference: [D9].

Essential theory and hypotheses

H0: mean_setosa = mean_versicolor. H1: mean_setosa != mean_versicolor. The test statistic measures the observed mean difference relative to its estimated standard error. Welch's t-test is used with equal_var=False because equal population variances need not be assumed. It still assumes independent observations and a suitable sampling model; the bundled educational data are not proof of a random sample from all flowers.

A two-sided p-value is the probability, under H0 and the model assumptions, of a test statistic at least as extreme in either direction. It is not the probability that H0 is true. Reject H0 when p<0.05; otherwise state insufficient evidence to reject it.

Step-by-step procedure

1. Extract the two species' sepal-length Series separately and check that both have 50 values.
2. State H0, H1 and alpha before interpreting the result.
3. Call stats.ttest_ind with equal_var=False and alternative="two-sided".
4. Display group means, t-statistic and p-value in scientific notation.
5. Compare the p-value with alpha and state a qualified conclusion.

Complete Python program

```
from scipy import stats
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
features = [
    "SepalLengthCm", "SepalWidthCm",
    "PetalLengthCm", "PetalWidthCm"
]
species = ["Iris-" + name for name in iris.target_names]
df = pd.DataFrame(iris.data, columns=features)
df["Species"] = pd.Categorical.from_codes(
    iris.target, categories=species
)

setosa = df.loc[df["Species"] == "Iris-setosa", "SepalLengthCm"]
versicolor = df.loc[
    df["Species"] == "Iris-versicolor", "SepalLengthCm"
]
alpha = 0.05
# Welch's independent two-sample t-test; no equal-variance assumption.
t_stat, p_value = stats.ttest_ind(
    setosa, versicolor, equal_var=False, alternative="two-sided"
)
print("H0: mean_setosa = mean_versicolor")
print("H1: mean_setosa != mean_versicolor")
print("Sample sizes:", len(setosa), len(versicolor))
print(f"Setosa mean: {setosa.mean():.6f} cm")
print(f"Versicolor mean: {versicolor.mean():.6f} cm")
print(f"t-statistic: {t_stat:.6f}")
print(f"p-value: {p_value:.6e}")
print("Significance level:", alpha)
if p_value < alpha:
    print("Reject H0: evidence of different population means.")
else:
    print("Do not reject H0: insufficient evidence of a difference.")
```

Code explanation

Boolean row selection isolates independent species groups. Welch's standard error uses the two group variances separately rather than pooling them. The order setosa minus versicolor determines the negative sign of t; swapping group order reverses that sign but leaves the two-sided p-value unchanged. `.6e` preserves a very small positive p-value that would look like 0.0000 with ordinary fixed rounding.

Verified output

```
H0: mean_setosa = mean_versicolor
H1: mean_setosa != mean_versicolor
Sample sizes: 50 50
Setosa mean: 5.006000 cm
Versicolor mean: 5.936000 cm
t-statistic: -10.520986
p-value: 3.746743e-17
Significance level: 0.05
Reject H0: evidence of different population means.
```

Observation

Setosa's sample mean is 5.006 cm and versicolor's is 5.936 cm, an observed difference of -0.930 cm. The two-sided p-value, about 3.75 x 10^-17, is well below 0.05. The negative t-statistic is consistent with setosa being the first argument and having the smaller mean. A tiny p-value is not an effect-size measure; the mean difference supplies useful magnitude context.

Result / examination-ready conclusion

At the 5% significance level, reject H0. Under the independent-sample assumptions, the data provide strong evidence that the population mean sepal lengths differ between setosa and versicolor. The observed setosa mean is lower. This does not prove H0 false with certainty or establish a causal explanation.

Common errors and troubleshooting

- All extracted values are missing: check that labels include the Iris- prefix used in this guide.
- equal_var omitted: SciPy defaults to a pooled Student test; explicitly choose Welch here. If an examiner specifically requests the pooled test, use equal_var=True and state its equal-variance assumption.
- p prints as 0.000000: use scientific notation; do not claim an exact zero probability.

Viva questions and answers

- Why use a two-sided test? The question asks whether means differ in either direction.
- What does equal_var=False request? Welch's independent t-test without equal population variances.
- Is p the probability H0 is true? No; it is computed under H0 and the test assumptions.
- If p>=0.05, can we say H0 is proved? No; we do not reject it for insufficient evidence of a difference.